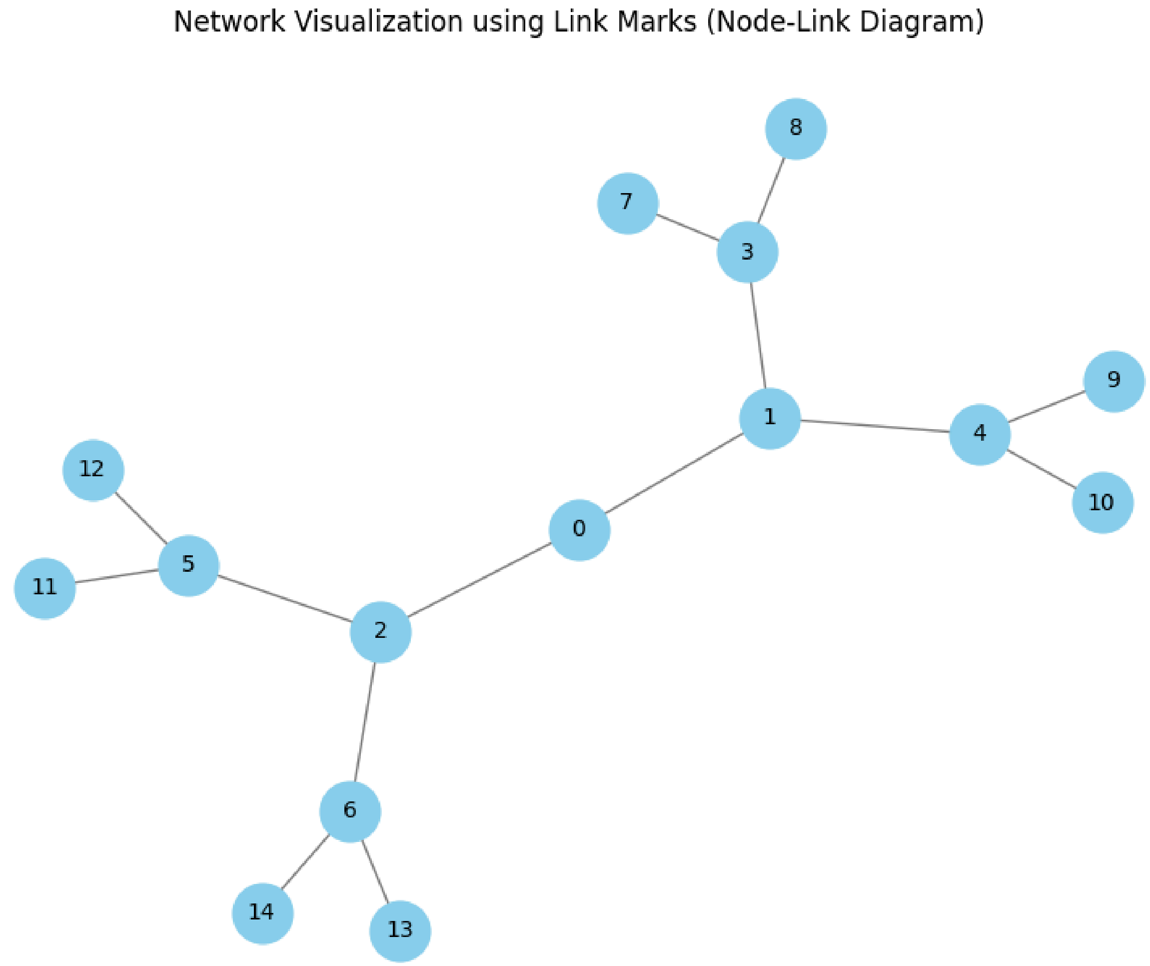
```python
import networkx as nx
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
```

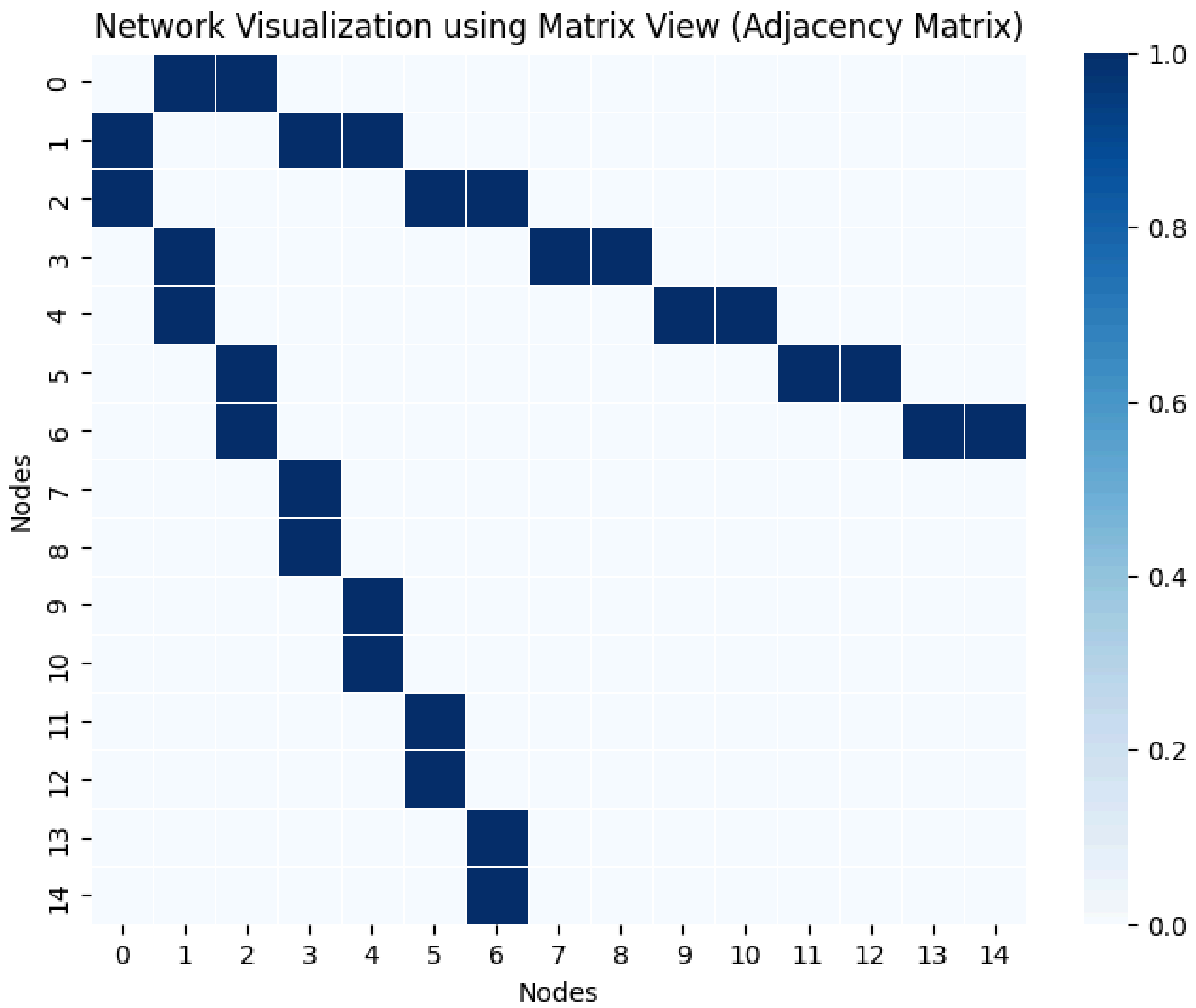
```python
G = nx.balanced_tree(r=2, h=3)
```

```python
plt.figure(figsize=(8,6))
pos = nx.spring_layout(G, seed=42)
nx.draw(G, pos, with_labels=True, node_color='skyblue', node_size=700,
        edge_color='gray', font_size=10)
plt.title("Network Visualization using Link Marks (Node-Link Diagram)")
plt.show()
```



```
adj_matrix = nx.to_numpy_array(G)
nodes = list(G.nodes)
df = pd.DataFrame(adj_matrix, index=nodes, columns=nodes)
```

```
plt.figure(figsize=(8,6))
sns.heatmap(df, cmap="Blues", linewidths=0.5, cbar=True)
plt.title("Network Visualization using Matrix View (Adjacency Matrix)")
plt.xlabel("Nodes")
plt.ylabel("Nodes")
plt.show()
```




```
print("Number of nodes:", G.number_of_nodes())
print("Number of edges:", G.number_of_edges())
print("Is the graph a tree?", nx.is_tree(G))
print("Average degree:", np.mean([d for n, d in G.degree()]))
```

```
Number of nodes: 15
Number of edges: 14
Is the graph a tree? True
Average degree: 1.8666666666666667
```